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Chapter 1

The Proofs of Vector Identities using Levi-Civita Tensor

1.1 Definitions

1.1.1 Einstein summation convention

Einstein (1916) introduced the following summation convention:

If an index appears exactly twice in a single term, it is understood that one should automatically sum over that index. Such an index is called a *dummy index*.

In this note, the indices i, j, k typically range from 1 to 3 in an orthogonal coordinate system, and we write:

$$\begin{aligned}\vec{V} &= \sum_{k=1}^3 V_k \hat{e}_k = V_k \hat{e}_k, & \vec{V} \cdot \vec{F} &= \sum_{k=1}^3 V_k F_k = V_k F_k \\ \nabla \varphi &= \hat{e}_k \partial_k \varphi = \hat{e}_k \frac{\partial \varphi}{\partial x_k}, & \nabla \cdot \vec{D} &= \partial_k D_k = \frac{\partial D_k}{\partial x_k} \\ C_{ik} &= \sum_{j=1}^M A_{ij} B_{jk} = A_{ij} B_{jk}\end{aligned}$$

Note that an index appearing only once is not a summation index.

1.1.2 Kronecker Delta and Levi-Civita Symbol

Definitions

The Roman indices i, j, k usually correspond to the x, y, z axes.

Kronecker delta:

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Levi-Civita symbol:

$$\epsilon_{ijk} = \begin{cases} +1 & (ijk) \in \{xyz, yzx, zxy\} \\ -1 & (ijk) \in \{xzy, yxz, zyx\} \\ 0 & \text{otherwise} \end{cases}$$

Useful Identities

$$\begin{aligned}\hat{e}_i \cdot \hat{e}_j &= \delta_{ij}, & \delta_{kk} &= 3 \\ \partial_k r_j &= \delta_{jk}, & V_k \delta_{kj} &= V_j \\ [\vec{V} \times \vec{F}]_i &= \epsilon_{ijk} V_j F_k, & [\nabla \times \vec{A}]_i &= \epsilon_{ijk} \partial_j A_k \\ \delta_{ij} \epsilon_{ijk} &= 0, & \epsilon_{ijk} \epsilon_{ijk} &= 6\end{aligned}$$

A particularly useful identity can be expressed as a single summation over the repeated index i :

$$\epsilon_{ijk} \epsilon_{ist} = \delta_{js} \delta_{kt} - \delta_{jt} \delta_{ks}$$

When no repeated indices appear, this identity can be generalized in determinant form:

$$\epsilon_{klt}\epsilon_{mpq} = \begin{vmatrix} \delta_{km} & \delta_{kp} & \delta_{kq} \\ \delta_{\ell m} & \delta_{\ell p} & \delta_{\ell q} \\ \delta_{tm} & \delta_{tp} & \delta_{tq} \end{vmatrix}$$

Useful Identities for 3×3 matrices

Now, let $\mathbf{C}$ be a 3×3 matrix with elements $C_{11}, C_{12}, \dots$. The determinant of this matrix can be expressed as a column expansion:

$$\det \mathbf{C} = \epsilon_{ijk} C_{1i} C_{j2} C_{k3}$$

Or equivalently, as a row expansion:

$$\det \mathbf{C} = \epsilon_{ijk} C_{i1} C_{2j} C_{3k}$$

The following identity related to the determinant of a 3×3 matrix $\mathbf{C}$ is especially useful:

$$\epsilon_{lmn} \det \mathbf{C} = \epsilon_{ijk} C_{li} C_{mj} C_{nk}$$

This equation compactly represents the component form of a determinant using the Levi-Civita symbol.

The left-hand side $\epsilon_{lmn} \det \mathbf{C}$ represents an antisymmetric tensor multiplied by the determinant, while the right-hand side expresses it as a contracted sum involving three Levi-Civita symbols and the matrix components.

In particular, the right-hand side expands as:

$$\epsilon_{ijk} C_{li} C_{mj} C_{nk} = C_{\ell 1} C_{m2} C_{n3} - C_{\ell 1} C_{m3} C_{n2} - C_{\ell 2} C_{m1} C_{n3} + C_{\ell 2} C_{m3} C_{n1} + C_{\ell 3} C_{m1} C_{n2} - C_{\ell 3} C_{m2} C_{n1}$$

This corresponds exactly to the classical cofactor expansion of a 3×3 determinant, demonstrating that the Levi-Civita symbol correctly encodes the sign of each term.

1.1.3 Example 1. Levi-Civita Basic Practice

Compute the values of the following expressions. (All expressions follow the Einstein summation convention.)

1. δ_{ii}
2. $\delta_{ij}\epsilon_{ijk}$
3. $\epsilon_{ijk}\epsilon_{ljk}$
4. $\epsilon_{ijk}\epsilon_{ijk}$

1.1.4 Example 2. Proving vector identities using Levi-Civita tensors

Prove the following vector identities.

1. $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$
2. $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})$
3. $\mathbf{A} \times [\mathbf{B} \times (\mathbf{C} \times \mathbf{D})] = \mathbf{B}[\mathbf{A} \cdot (\mathbf{C} \times \mathbf{D})] - (\mathbf{A} \cdot \mathbf{B})(\mathbf{C} \times \mathbf{D})$

Solution for Example 1

1. $\delta_{ii} = \delta_{xx} + \delta_{yy} + \delta_{zz} = 1 + 1 + 1 = 3$
2. $\delta_{ij}\epsilon_{ijk} = \epsilon_{iik} = 0$
3. $\epsilon_{ijk}\epsilon_{ljk} = \epsilon_{jki}\epsilon_{jkl} = \delta_{kk}\delta_{il} - \delta_{kl}\delta_{ik} = \delta_{il} - \delta_{il} = 0$
4. $\epsilon_{ijk}\epsilon_{ijk} = \delta_{jj}\delta_{kk} - \delta_{jk}\delta_{kj} = 3 \cdot 3 - \delta_{jj} = 9 - 3 = 6$

Solution for Example 2

1.

$$\begin{aligned}
[\mathbf{A} \times (\mathbf{B} \times \mathbf{C})]_i &= \varepsilon_{ijk} A_j (\mathbf{B} \times \mathbf{C})_k \\
&= \varepsilon_{ijk} A_j \varepsilon_{klm} B_l C_m \\
&= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) A_j B_l C_m \\
&= A_j B_i C_j - A_j B_j C_i \\
&= B_i (A_j C_j) - C_i (A_j B_j) \\
&= [\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})]_i
\end{aligned}$$

2.

$$\begin{aligned}
(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) &= \varepsilon_{ijk} A_j B_k \cdot \varepsilon_{ilm} C_l D_m \\
&= (\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}) A_j B_k C_l D_m \\
&= A_j B_k C_j D_k - A_j B_k C_k D_j \\
&= (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})
\end{aligned}$$

3.

$$\begin{aligned}
\mathbf{B} \times (\mathbf{C} \times \mathbf{D}) &= \mathbf{C}(\mathbf{B} \cdot \mathbf{D}) - \mathbf{D}(\mathbf{B} \cdot \mathbf{C}) \quad (\text{벡터 삼중곱 항등식}) \\
\mathbf{A} \times [\mathbf{B} \times (\mathbf{C} \times \mathbf{D})] &= \mathbf{A} \times [\mathbf{C}(\mathbf{B} \cdot \mathbf{D}) - \mathbf{D}(\mathbf{B} \cdot \mathbf{C})] \\
&= (\mathbf{B} \cdot \mathbf{D})\mathbf{A} \times \mathbf{C} - (\mathbf{B} \cdot \mathbf{C})\mathbf{A} \times \mathbf{D} \\
&= \mathbf{B}[\mathbf{A} \cdot (\mathbf{C} \times \mathbf{D})] - (\mathbf{A} \cdot \mathbf{B})(\mathbf{C} \times \mathbf{D})
\end{aligned}$$

또는

$$\begin{aligned}
[\mathbf{A} \times (\mathbf{B} \times (\mathbf{C} \times \mathbf{D}))]_i &= \varepsilon_{ijk} A_j (\mathbf{B} \times (\mathbf{C} \times \mathbf{D}))_k \\
&= \varepsilon_{ijk} A_j \varepsilon_{klm} B_l (\mathbf{C} \times \mathbf{D})_m \\
&= \varepsilon_{ijk} \varepsilon_{klm} A_j B_l \varepsilon_{mnp} C_n D_p \\
&= (\delta_{il}\delta_{jn} - \delta_{in}\delta_{jl}) A_j B_l \varepsilon_{mnp} C_n D_p \\
&= A_j B_i \varepsilon_{jnp} C_n D_p - A_j B_j \varepsilon_{inp} C_n D_p \\
&= B_i (A_j \varepsilon_{jnp} C_n D_p) - (A_j B_j) \varepsilon_{inp} C_n D_p \\
&= [\mathbf{B}(\mathbf{A} \cdot (\mathbf{C} \times \mathbf{D}))]_i - [(\mathbf{A} \cdot \mathbf{B})(\mathbf{C} \times \mathbf{D})]_i
\end{aligned}$$

$$\therefore \mathbf{A} \times [\mathbf{B} \times (\mathbf{C} \times \mathbf{D})] = \mathbf{B}[\mathbf{A} \cdot (\mathbf{C} \times \mathbf{D})] - (\mathbf{A} \cdot \mathbf{B})(\mathbf{C} \times \mathbf{D})$$

1.1.5 Example 3. Decomposition identity

Prove the following vector identities.

1. Let $\mathbf{b}$ be a vector and $\hat{n}$ be a unit vector. Prove that $\mathbf{b} = (\mathbf{b} \cdot \hat{n})\hat{n} + \hat{n} \times \mathbf{b} \times \hat{n}$ and interpret this identity geometrically.
2. Consider the $\hbar = 1$ version of the quantum mechanical angular momentum operator, $\mathbf{L} = -i\mathbf{r} \times \nabla$. Prove the following radial-angular decomposition identity for the gradient operator:

$$\nabla = \hat{r} \frac{\partial}{\partial r} - \frac{i}{r} \hat{r} \times \mathbf{L}$$

3. $A_i B_j = \frac{1}{2} \epsilon_{ijk} (A \times B)_k + \frac{1}{2} (A_i B_j + A_j B_i)$

Solution for Example 3

1.

$$\begin{aligned}
b_j &= (b_i n_i) n_j + b_j - (b_i n_i) n_j \\
&= (b_i n_i) n_j + \{(n_i n_i) b_j - (b_i n_i) n_j\}
\end{aligned}$$

On the other hand,

$$\begin{aligned}
(\hat{n} \times \mathbf{b} \times \hat{n})_j &= \epsilon_{j p q} n_p (\mathbf{b} \times \hat{n})_q \\
&= \epsilon_{j p q} n_p (\epsilon_{q r s} b_r n_s) \\
&= \epsilon_{q j p} \epsilon_{q r s} (n_p b_r n_s) \\
&= (\delta_{j r} \delta_{p s} - \delta_{j s} \delta_{p r}) (n_p b_r n_s) \\
&= (n_p n_p) b_j - (n_p b_p) n_j
\end{aligned}$$

Therefore,

$$\begin{aligned}
b_j &= (b_i n_i) n_j + (n_i n_i) b_j - (b_i n_i) n_j \\
&= (b_i n_i) n_j + (\hat{n} \times \mathbf{b} \times \hat{n})_j \\
\therefore \mathbf{b} &= (\mathbf{b} \cdot \hat{n}) \hat{n} + \hat{n} \times \mathbf{b} \times \hat{n} \\
&\equiv b_{\parallel} + b_{\perp}
\end{aligned}$$

2. Setting the $\hat{n} = \hat{r}$, $\mathbf{b} = \nabla$ in the decomposition identity in the above,

$$\begin{aligned}
\nabla &= \hat{r}(\hat{r} \cdot \nabla) + \hat{r} \times (\nabla \times \hat{r}) \\
&= \hat{r} \frac{\partial}{\partial r} - \hat{r} \times (\hat{r} \times \nabla) \\
&= \hat{r} \frac{\partial}{\partial r} - i \hat{r} \times (-i) \left(\frac{\mathbf{r}}{r} \times \nabla \right) \\
&= \hat{r} \frac{\partial}{\partial r} - \frac{i}{r} \hat{r} \times (-i) (\mathbf{r} \times \nabla) \\
&= \hat{r} \frac{\partial}{\partial r} - \frac{i}{r} \hat{r} \times \mathbf{L}
\end{aligned}$$

3.

$$A_i B_j = \frac{1}{2} (A_i B_j + A_j B_i) + \frac{1}{2} (A_i B_j - A_j B_i)$$

Now, note that

$$\begin{aligned}
\epsilon_{i j k} (A \times B)_k &= \epsilon_{i j k} \epsilon_{k p q} A_p B_q \\
&= \epsilon_{k i j} \epsilon_{k p q} A_p B_q \\
&= (\delta_{i p} \delta_{j q} - \delta_{i q} \delta_{j p}) A_p B_q \\
&= A_i B_j - A_j B_i
\end{aligned}$$

Hence,

$$\begin{aligned}
A_i B_j &= \frac{1}{2} (A_i B_j + A_j B_i) + \frac{1}{2} (A_i B_j - A_j B_i) \\
&= \frac{1}{2} (A_i B_j + A_j B_i) + \frac{1}{2} \epsilon_{i j k} (A \times B)_k
\end{aligned}$$

Example 4. Identities in Quantum Mechanics

Prove the following identities.

1. In quantum mechanics, the Cartesian components of the angular momentum operator $\hat{\mathbf{L}}$ obey the commutation relation $[\hat{L}_i, \hat{L}_j] = \epsilon_{ijk} \hat{L}_k$. Let $\mathbf{a}$ and $\mathbf{b}$ be constant vectors and prove the commutator identity

$$[\hat{\mathbf{L}} \cdot \mathbf{a}, \hat{\mathbf{L}} \cdot \mathbf{b}] = i\hbar \hat{\mathbf{L}} \cdot (\mathbf{a} \times \mathbf{b})$$

2. The 2×2 Pauli matrices σ_x, σ_y and σ_z used in quantum mechanics satisfy $\sigma_i \sigma_j = \delta_{ij} + i\epsilon_{ijk} \sigma_k$. If $\mathbf{a}$ and $\mathbf{b}$ are ordinary vectors, prove that

$$(\boldsymbol{\sigma} \cdot \mathbf{a})(\boldsymbol{\sigma} \cdot \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} + i \boldsymbol{\sigma} \cdot (\mathbf{a} \times \mathbf{b})$$

Solution for Example 4

1.

$$\begin{aligned}
\hat{\mathbf{L}} \cdot \mathbf{a} &= a_i \hat{L}_i, & \hat{\mathbf{L}} \cdot \mathbf{b} &= b_j \hat{L}_j \\
\Rightarrow [\hat{\mathbf{L}} \cdot \mathbf{a}, \hat{\mathbf{L}} \cdot \mathbf{b}] &= [a_i \hat{L}_i, b_j \hat{L}_j] = a_i b_j [\hat{L}_i, \hat{L}_j] \\
&= a_i b_j (i\hbar \varepsilon_{ijk} \hat{L}_k) \\
&= i\hbar \varepsilon_{ijk} a_i b_j \hat{L}_k = i\hbar (\mathbf{a} \times \mathbf{b})_k \hat{L}_k \\
&= i\hbar \hat{\mathbf{L}} \cdot (\mathbf{a} \times \mathbf{b})
\end{aligned}$$

$$\therefore [\hat{\mathbf{L}} \cdot \mathbf{a}, \hat{\mathbf{L}} \cdot \mathbf{b}] = i\hbar \hat{\mathbf{L}} \cdot (\mathbf{a} \times \mathbf{b})$$

2.

$$\begin{aligned}
(\boldsymbol{\sigma} \cdot \mathbf{a})(\boldsymbol{\sigma} \cdot \mathbf{b}) &= a_i b_j \sigma_i \sigma_j \\
&= a_i b_j (\delta_{ij} I + i\varepsilon_{ijk} \sigma_k) \\
&= a_i b_i I + i a_i b_j \varepsilon_{ijk} \sigma_k \\
&= (\mathbf{a} \cdot \mathbf{b}) I + i(\mathbf{a} \times \mathbf{b})_k \sigma_k \\
&= \mathbf{a} \cdot \mathbf{b} + i\boldsymbol{\sigma} \cdot (\mathbf{a} \times \mathbf{b})
\end{aligned}$$

$$\therefore (\boldsymbol{\sigma} \cdot \mathbf{a})(\boldsymbol{\sigma} \cdot \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} + i\boldsymbol{\sigma} \cdot (\mathbf{a} \times \mathbf{b})$$

1.2 Product Rules for Vector Calculus

There are two ways to construct a scalar as the product of two functions:

$$\begin{aligned} fg & \text{ (product of two scalar functions),} \\ \mathbf{A} \cdot \mathbf{B} & \text{ (dot product of two vector functions),} \end{aligned}$$

and two ways to construct a vector:

$$\begin{aligned} f\mathbf{A} & \text{ (scalar times vector),} \\ \mathbf{A} \times \mathbf{B} & \text{ (cross product of two vectors).} \end{aligned}$$

Accordingly, there are six product rules:

(i) Gradient of scalar product:

$$\nabla(fg) = f\nabla g + g\nabla f$$

(ii) Gradient of dot product:

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$$

(iii) Divergence of scalar times vector:

$$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$$

(iv) Divergence of cross product:

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

(v) Curl of scalar times vector:

$$\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$$

(vi) Curl of cross product:

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$$

1.2.1 Example 4. Proof of the product rules in Vector calculus

Prove the following vector identities.

1. $\nabla(fg) = f\nabla g + g\nabla f$
2. $\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$
3. $\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$
4. $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
5. $\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$
6. $\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$

Solution for Example 5

1. skipped
2. 좌변 계산:

스칼라 함수 $\mathbf{A} \cdot \mathbf{B} = A_j B_j$ 에 대해 그라디언트는

$$[\nabla(\mathbf{A} \cdot \mathbf{B})]_i = \partial_i(A_j B_j) = (\partial_i A_j) B_j + A_j (\partial_i B_j)$$

우변 항들 계산:

(a) $[\mathbf{A} \times (\nabla \times \mathbf{B})]_i$ 부터 보자:

$$[\nabla \times \mathbf{B}]_k = \epsilon_{klm} \partial_l B_m \Rightarrow [\mathbf{A} \times (\nabla \times \mathbf{B})]_i = \epsilon_{ijk} A_j (\nabla \times \mathbf{B})_k = \epsilon_{ijk} A_j \epsilon_{klm} \partial_l B_m$$

Levi-Civita 항등식($\epsilon_{ijk} \epsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}$) 을 사용하면

$$\epsilon_{ijk} \epsilon_{klm} A_j \partial_l B_m = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) A_j \partial_l B_m = A_m \partial_i B_m - A_j \partial_j B_i$$

(b) $[\mathbf{B} \times (\nabla \times \mathbf{A})]_i$ 도 같은 방식으로

$$[\nabla \times \mathbf{A}]_k = \epsilon_{klm} \partial_l A_m \Rightarrow [\mathbf{B} \times (\nabla \times \mathbf{A})]_i = \epsilon_{ijk} B_j \epsilon_{klm} \partial_l A_m$$

따라서

$$\epsilon_{ijk} \epsilon_{klm} B_j \partial_l A_m = B_m \partial_i A_m - B_j \partial_j A_i$$

$$(c) [(\mathbf{A} \cdot \nabla) \mathbf{B}]_i = A_j \partial_j B_i, \quad [(\mathbf{B} \cdot \nabla) \mathbf{A}]_i = B_j \partial_j A_i$$

이제 위 네 항을 모두 더하면:

$$[\text{우변}]_i = A_m \partial_i B_m - A_j \partial_j B_i + B_m \partial_i A_m - B_j \partial_j A_i + A_j \partial_j B_i + B_j \partial_j A_i$$

$$= A_m \partial_i B_m + B_m \partial_i A_m = \partial_i(A_m B_m) = \partial_i(\mathbf{A} \cdot \mathbf{B})$$

이는 좌변 $[\nabla(\mathbf{A} \cdot \mathbf{B})]_i$ 와 같으므로 항등식이 증명됨.

3. $\partial_i(f A_i) = f \partial_i A_i + A_i \partial_i f$
4. $\partial_i(\epsilon_{ijk} A_j B_k) = \epsilon_{ijk} (\partial_i A_j) B_k + \epsilon_{ijk} A_j (\partial_i B_k) = (\nabla \times \mathbf{A}) \cdot \mathbf{B} - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
5. $(\nabla \times (f \mathbf{A}))_i = \epsilon_{ijk} \partial_j (f A_k) = \epsilon_{ijk} [A_k \partial_j f + f \partial_j A_k] = f(\nabla \times \mathbf{A})_i - (\mathbf{A} \times \nabla f)_i$
- 6.

$$\begin{aligned} [\nabla \times (\mathbf{A} \times \mathbf{B})]_i &= \epsilon_{ijk} \partial_j (\epsilon_{klm} A_l B_m) \\ &= \epsilon_{ijk} \epsilon_{klm} \partial_j (A_l B_m) \\ &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \partial_j (A_l B_m) \\ &= \partial_j A_i B_j + A_i \partial_j B_j - \partial_j A_j B_i - A_j \partial_j B_i \end{aligned}$$

정리하면:

$$(\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$$

1.3 Second Derivatives

The gradient, divergence, and curl are the only first-order vector differential operators we can construct using ∇ . By applying ∇ twice, we can generate five distinct second-order combinations, depending on whether the input is a scalar or vector field.

First, for a scalar field T , we can take:

- the **divergence of the gradient**: $\nabla \cdot (\nabla T) = \nabla^2 T$, which is the Laplacian of T ,
- the **curl of the gradient**: $\nabla \times (\nabla T) = \mathbf{0}$, which always vanishes.

Second, for a vector field $\mathbf{v}$, we can consider:

- the **gradient of the divergence**: $\nabla(\nabla \cdot \mathbf{v})$, which is well-defined but not commonly used; it is simply the gradient of the scalar field $\nabla \cdot \mathbf{v}$,
- the **divergence of the curl**: $\nabla \cdot (\nabla \times \mathbf{v}) = 0$, which vanishes identically due to the antisymmetry of the Levi-Civita symbol,
- the **curl of the curl**:

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v},$$

which expands into a combination of the gradient of divergence and the vector Laplacian.

This exhausts all second-order combinations involving ∇ . Not all of these give rise to new structures: two of them vanish identically, and the curl of a curl reduces to a combination of the other two. Therefore, we conclude that there are fundamentally only two types of second derivatives:

- the Laplacian: $\nabla^2 T$ or $\nabla^2 \mathbf{v}$,
- the gradient of divergence: $\nabla(\nabla \cdot \mathbf{v})$.

All other second derivatives either vanish or are reducible to these two, which suffices for virtually all physical applications. This shows the unifying power of the operator ∇ : if one deeply understands its structure and vectorial nature, the rest of vector calculus flows naturally from it.

1.3.1 Example 5. Vector identities for second derivatives

Prove the following vector identities.

1. $\nabla \times (\nabla f) = 0$
2. $\nabla \cdot (\nabla \times \mathbf{g}) = 0$
3. $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$

Solution for Example 6

1.

$$\begin{aligned}
[\nabla \times (\nabla f)]_i &= \epsilon_{ijk} \partial_j \partial_k f \\
&= \frac{1}{2} (\epsilon_{ijk} + \epsilon_{ikj}) \partial_j \partial_k f \\
&= \frac{1}{2} (\epsilon_{ijk} + (-\epsilon_{ijk})) \partial_j \partial_k f \\
&= \frac{1}{2} (0) \cdot \partial_j \partial_k f = 0
\end{aligned}$$

2.

$$\begin{aligned}
(\nabla \cdot (\nabla \times \mathbf{g})) &= \partial_i (\epsilon_{ijk} \partial_j g_k) \\
&= \frac{1}{2} \partial_i \partial_j g_k \epsilon_{ijk} + \frac{1}{2} \partial_j \partial_i g_k \epsilon_{jik} \\
&= \frac{1}{2} (\epsilon_{ijk} + \epsilon_{jik}) \partial_i \partial_j g_k = 0 \quad (\text{since } \epsilon_{ijk} = -\epsilon_{jik})
\end{aligned}$$

3.

$$\begin{aligned}
[\nabla \times (\nabla \times \mathbf{A})]_i &= \epsilon_{ijk} \partial_j (\nabla \times \mathbf{A})_k \\
&= \epsilon_{ijk} \partial_j (\epsilon_{kpq} \partial_p A_q) \\
&= \epsilon_{ijk} \epsilon_{kpq} \partial_j \partial_p A_q \\
&= (\delta_{ip} \delta_{jq} - \delta_{iq} \delta_{jp}) \partial_j \partial_p A_q \\
&= \partial_j \partial_i A_j - \partial_j \partial_j A_i \\
&= \nabla_i (\nabla \cdot \mathbf{A}) - \nabla^2 A_i
\end{aligned}$$

 $\cong$,

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

1.3.2 Example 6. Vector identities regarding the position vector

All functions of $\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$ are assumed to be differentiable. Prove the following identities.

1. $\nabla r = \hat{\mathbf{r}}$, where $r = \sqrt{x^2 + y^2 + z^2}$
2. $\nabla \times \mathbf{r} = 0$
3. $\nabla f(r) = f'(r)\hat{\mathbf{r}}$, where $f'(r) = \frac{df}{dr}$
4. $\nabla^2 f(r) = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{df}{dr} \right)$
5. $\nabla \cdot (f\mathbf{r}) = \frac{d}{dr} (r^3 f) \cdot \frac{1}{r^2}$
6. $\nabla \times (f\mathbf{r}) = 0$
7. $\nabla \cdot \mathbf{g}(r) = \mathbf{g}' \cdot \hat{\mathbf{r}}$
8. $\nabla \times \mathbf{g}(r) = \hat{\mathbf{r}} \times \mathbf{g}'$
9. $(\mathbf{g} \cdot \nabla)\mathbf{r} = \mathbf{g}$
10. $(\mathbf{r} \cdot \nabla)\mathbf{g} = r\mathbf{g}'$
11. $\nabla(\mathbf{r} \cdot \mathbf{g}) = \mathbf{g} + \frac{(\mathbf{r} \cdot \mathbf{g}')\mathbf{r}}{r}$
12. $\nabla \times (\mathbf{g} \times \mathbf{r}) = 2\mathbf{g} + r\mathbf{g}' - \frac{(\mathbf{r} \cdot \mathbf{g}')\mathbf{r}}{r}$
13. $\nabla \cdot (\mathbf{g} \times \mathbf{r}) = 0$
14. $\nabla(\mathbf{c} \cdot \mathbf{r}) = \mathbf{c}$, where $\mathbf{c}$ is constant

Solution for Example 7

1.

$$r = \sqrt{x_i x_i}, \quad \partial_j r = \frac{1}{2\sqrt{x_k x_k}} \cdot \partial_j (x_i x_i) = \frac{x_j}{r} \quad (\because \partial_j x_i = \delta_{ij})$$

$$\Rightarrow \nabla r = \left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r} \right) = \hat{r}$$

2.

$$(\nabla \times \mathbf{r})_i = \varepsilon_{ijk} \partial_j r_k = \varepsilon_{ijk} \delta_{jk} = 0 \quad (\because \varepsilon_{ijk} \text{ is zero if } j = k)$$

3.

$$\partial_j f(r) = \frac{df}{dr} \cdot \partial_j r = f'(r) \cdot \frac{x_j}{r} \quad (\because \partial_j r = \frac{x_j}{r})$$

$$\Rightarrow \nabla f(r) = f'(r) \cdot \hat{r}$$

4.

$$\nabla^2 f(r) = \partial_i \partial_i f(r) = \partial_i \left(f'(r) \frac{x_i}{r} \right) = f''(r) \frac{x_i x_i}{r^2} + f'(r) \cdot \partial_i \left(\frac{x_i}{r} \right)$$

$$= f''(r) + f'(r) \cdot \left(\frac{3r^2 - x_i x_i}{r^3} \right) = f''(r) + \frac{2}{r} f'(r) \quad (\because \partial_i x_i = 3, \quad x_i x_i = r^2)$$

5.

$$\nabla \cdot (f\mathbf{r}) = \partial_i (f x_i) = f \partial_i x_i + x_i \partial_i f = 3f + \frac{x_i x_i}{r} f'(r) = 3f + r f'(r)$$

$$= \frac{1}{r^2} \cdot \frac{d}{dr} (r^3 f) \quad (\because \frac{d}{dr} (r^3 f) = 3r^2 f + r^3 f')$$

6.

$$\nabla \times (f\mathbf{r})_i = \varepsilon_{ijk} \partial_j (f x_k) = \varepsilon_{ijk} (\partial_j f \cdot x_k + f \cdot \partial_j x_k) = \varepsilon_{ijk} \left(\frac{x_j}{r} f'(r) x_k + f \delta_{jk} \right)$$

$$= \varepsilon_{ijk} \left(\frac{f'(r)}{r} x_j x_k + f \delta_{jk} \right) \Rightarrow \varepsilon_{ijk} x_j x_k = 0, \quad \varepsilon_{ijk} \delta_{jk} = 0 \Rightarrow \nabla \times (f\mathbf{r}) = 0$$

7.

$$\nabla \cdot \mathbf{g}(r) = \partial_i g_i(r) = \frac{dg_i}{dr} \cdot \frac{\partial r}{\partial x_i} = g'_i(r) \cdot \frac{x_i}{r} = \mathbf{g}'(r) \cdot \hat{r}$$

8.

$$(\nabla \times \mathbf{g}(r))_i = \varepsilon_{ijk} \partial_j g_k(r) = \varepsilon_{ijk} \frac{dg_k}{dr} \cdot \frac{x_j}{r} = (\hat{r} \times \mathbf{g}'(r))_i \Rightarrow \nabla \times \mathbf{g}(r) = \hat{r} \times \mathbf{g}'(r)$$

9.

$$[(\mathbf{g} \cdot \nabla) \mathbf{r}]_i = g_j \partial_j x_i = g_j \delta_{ji} = g_i \Rightarrow (\mathbf{g} \cdot \nabla) \mathbf{r} = \mathbf{g}$$

10.

$$[(\mathbf{r} \cdot \nabla) \mathbf{g}]_i = x_j \partial_j g_i(r) = x_j \cdot \frac{dg_i}{dr} \cdot \frac{\partial r}{\partial x_j} = x_j \cdot g'_i(r) \cdot \frac{x_j}{r} = g'_i(r) \cdot \frac{x_j x_j}{r} = r g'_i(r) \Rightarrow (\mathbf{r} \cdot \nabla) \mathbf{g} = r \mathbf{g}'(r)$$

11.

$$\begin{aligned}\nabla(\mathbf{r} \cdot \mathbf{g}) &= \partial_i(x_j g_j) = \delta_{ij} g_j + x_j \partial_i g_j = g_i + \partial_i g_j \cdot x_j \\ &= \mathbf{g} + \nabla g_j \cdot x_j = \mathbf{g} + \frac{x_j}{r} g'_j(r) = \mathbf{g} + \frac{(\mathbf{r} \cdot \mathbf{g}')}{r} \hat{r}\end{aligned}$$

12.

$$\begin{aligned}[\nabla \times (\mathbf{g} \times \mathbf{r})]_i &= \varepsilon_{ijk} \partial_j (\varepsilon_{klm} g_l r_m) = \varepsilon_{ijk} \varepsilon_{klm} \partial_j (g_l x_m) \\ &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \partial_j (g_l x_m) = \partial_j (g_i x_j) - \partial_j (g_j x_i) \\ &= x_j \partial_j g_i + g_i \delta_{jj} - x_i \partial_j g_j - g_j \delta_{ji} = (\mathbf{r} \cdot \nabla) \mathbf{g} + 3\mathbf{g} - \nabla(\mathbf{r} \cdot \mathbf{g}) \\ &= r\mathbf{g}' + 3\mathbf{g} - \left(\mathbf{g} + \frac{(\mathbf{r} \cdot \mathbf{g}')}{r} \hat{r} \right) = 2\mathbf{g} + r\mathbf{g}' - \frac{(\mathbf{r} \cdot \mathbf{g}')}{r} \hat{r}\end{aligned}$$

13.

$$\begin{aligned}\nabla \cdot (\mathbf{g} \times \mathbf{r}) &= \partial_i (\varepsilon_{ijk} g_j x_k) = \varepsilon_{ijk} \partial_i (g_j x_k) = \varepsilon_{ijk} (\partial_i g_j x_k + g_j \partial_i x_k) \\ &= \varepsilon_{ijk} \left(\frac{dg_j}{dr} \cdot \frac{x_i}{r} x_k + g_j \delta_{ik} \right) \Rightarrow \varepsilon_{ijk} x_i x_k = 0, \quad \varepsilon_{ijk} \delta_{ik} = 0 \Rightarrow \nabla \cdot (\mathbf{g} \times \mathbf{r}) = 0\end{aligned}$$

14.

$$\nabla \cdot (\mathbf{c} \cdot \mathbf{r}) = \partial_i (c_j x_j) = c_j \partial_i x_j = c_j \delta_{ij} = c_i \Rightarrow \nabla(\mathbf{c} \cdot \mathbf{r}) = \mathbf{c} \quad (\because \mathbf{c} = \text{const.})$$

1.4 Further applications of Vector differential operators

1.4.1 Functions of $\mathbf{r} - \mathbf{r}'$

Let $\mathbf{R} = \mathbf{r} - \mathbf{r}' = (x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}$. Then,

$$\nabla f(R) = f'(R)\hat{\mathbf{R}}, \quad \nabla \cdot \mathbf{g}(R) = \mathbf{g}'(R) \cdot \hat{\mathbf{R}}, \quad \nabla \times \mathbf{g}(R) = \hat{\mathbf{R}} \times \mathbf{g}'(R)$$

Moreover, because

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}, \quad \nabla' = \hat{\mathbf{x}} \frac{\partial}{\partial x'} + \hat{\mathbf{y}} \frac{\partial}{\partial y'} + \hat{\mathbf{z}} \frac{\partial}{\partial z'},$$

it is straightforward to confirm that

$$\nabla' f(R) = -\nabla f(R)$$

1.4.2 The Convective Derivative

Let $\phi(\mathbf{r}, t)$ be a scalar function of space and time. The total time derivative (convective derivative) of ϕ along a moving trajectory $\mathbf{r}(t)$, where $\mathbf{v}(t) = \dot{\mathbf{r}}(t)$, is

$$\frac{d\phi}{dt} = \frac{\partial \phi}{\partial t} + \frac{dx}{dt} \frac{\partial \phi}{\partial x} + \frac{dy}{dt} \frac{\partial \phi}{\partial y} + \frac{dz}{dt} \frac{\partial \phi}{\partial z} = \frac{\partial \phi}{\partial t} + (\mathbf{v} \cdot \nabla) \phi$$

For a vector function $\mathbf{g}(\mathbf{r}, t)$, the corresponding convective derivative is

$$\frac{d\mathbf{g}}{dt} = \frac{\partial \mathbf{g}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{g}$$

1.4.3 Taylor's Theorem

Taylor's theorem in one dimension is

$$f(x) = f(a) + (x - a) \left. \frac{df}{dx} \right|_{x=a} + \frac{1}{2!} (x - a)^2 \left. \frac{d^2 f}{dx^2} \right|_{x=a} + \dots$$

If $x \rightarrow x + \epsilon$ and $a \rightarrow x$, then

$$f(x + \epsilon) = f(x) + \epsilon \frac{df}{dx} + \frac{1}{2!} \epsilon^2 \frac{d^2 f}{dx^2} + \dots$$

Equivalently,

$$f(x + \epsilon) = \left[1 + \epsilon \frac{d}{dx} + \frac{1}{2!} \left(\epsilon \frac{d}{dx} \right)^2 + \dots \right] f(x) = \exp \left(\epsilon \frac{d}{dx} \right) f(x)$$

This generalizes for a function of three variables to

$$f(x + \epsilon_x, y + \epsilon_y, z + \epsilon_z) = \exp \left(\epsilon_x \frac{\partial}{\partial x} \right) \exp \left(\epsilon_y \frac{\partial}{\partial y} \right) \exp \left(\epsilon_z \frac{\partial}{\partial z} \right) f(x, y, z)$$

or

$$f(\mathbf{r} + \boldsymbol{\epsilon}) = \exp(\boldsymbol{\epsilon} \cdot \nabla) f(\mathbf{r}) = \left[1 + \boldsymbol{\epsilon} \cdot \nabla + \frac{1}{2!} (\boldsymbol{\epsilon} \cdot \nabla)^2 + \dots \right] f(\mathbf{r})$$